A note on a recent attack against SPEEDY-7-192

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Abstract. Recently, two independent differential attacks on SPEEDY-7-192 were proposed by Boura et al. [3] and by Beyne and Neyt [1]. Both works present, for the first time, a valid differential attack on SPEEDY-7-192 with time complexities of $2^{186.36}$ and 2^{185} respectively. In this note, by extending the search space for 1-round trails, we propose a new differential attack on SPEEDY-7-192 with both data and time complexity of $2^{174.80}$. This improves upon both previous attacks by more than a factor of 2^{10}.

1 Introduction

Shortly after the publication of our new differential attacks on the SPEEDY family of block ciphers [3], Beyne and Neyt also proposed a differential attack on the full version of SPEEDY-7-192 [1], with a slightly better complexity. Since it is currently computationally infeasible to directly search for the best differential characteristics against SPEEDY, both works rely on the same strategy to find *good* characteristics: first, search for 1-round trails, then chain them together. The set used by Beyne and Neyt shares many 1-round trails with the one used in [3]. However, some of these trails — particularly those with few active rows around the MixColumns layer but with probabilities lower than 2^{-46} — were discarded in our original search [3] due to efficiency constraints. After reviewing the work in [1], we incorporated these additional trails into our search space and re-ran the suite of tools we developed to identify the best differential attack against SPEEDY.

2 Distinguisher

As a result, we were indeed able to recover the attack presented by Beyne and Neyt in [1]. However, by exhaustively exploring all 5-round characteristics along with their best probabilistic extensions using our automatic approach as described in [3, Section 4.1], we discovered a significantly more efficient attack. This new attack is based on the main characteristic shown in Figure 1 that has probability $2^{-159.565}$. By clustering characteristics with the same input and output differences, we derived the differential attack illustrated in Figure 2, based on a differential of probability $2^{-158.632}$.

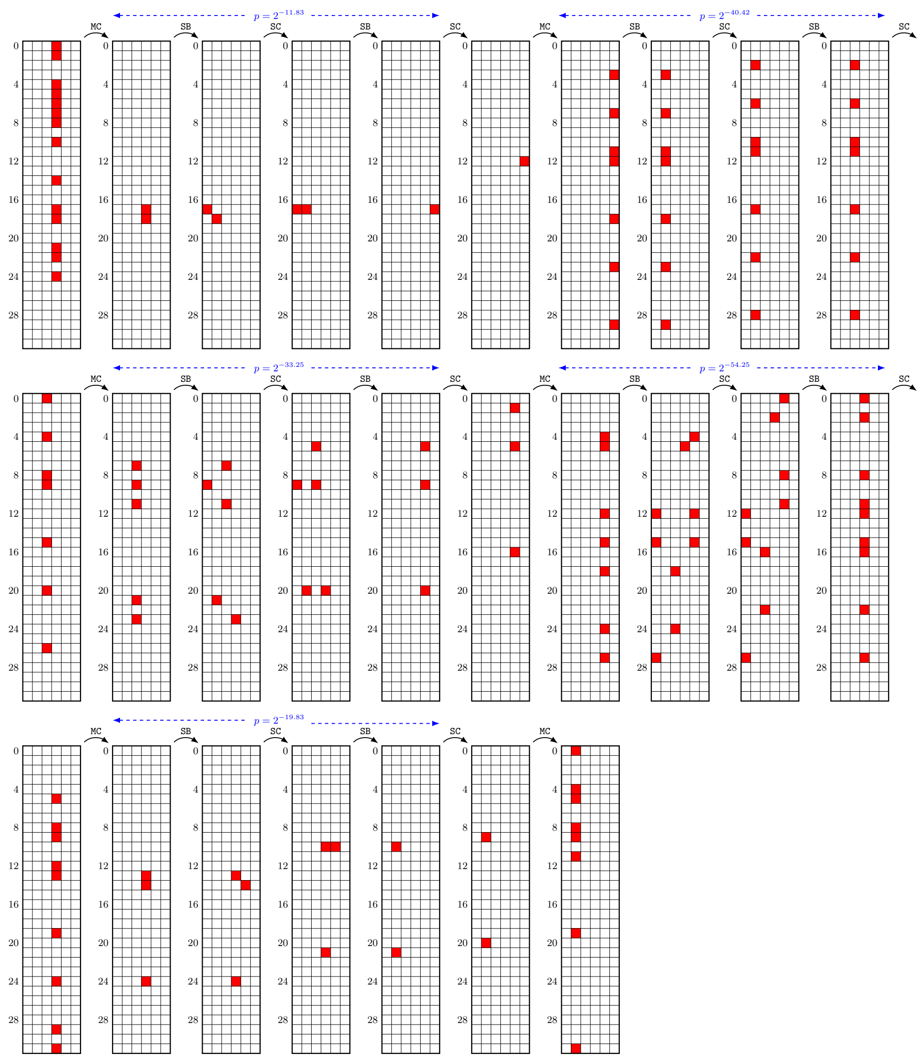

Fig. 1: 5-round dominant trail of the distinguisher used for the attack against `SPEEDY-7` of probability $2^{-159.565}$.

It is interesting to note that the distinguisher we use is far from optimal in terms of probability. There are in fact more than 4,000 5-round differential distinguishers with higher probabilities. However, it turns out to give the best full attack once the probabilistic extension is taken into account. This highlights the value of our systematic approach, which was an important contribution of our work and which we believe should be generalized to other targets.

3 Key recovery

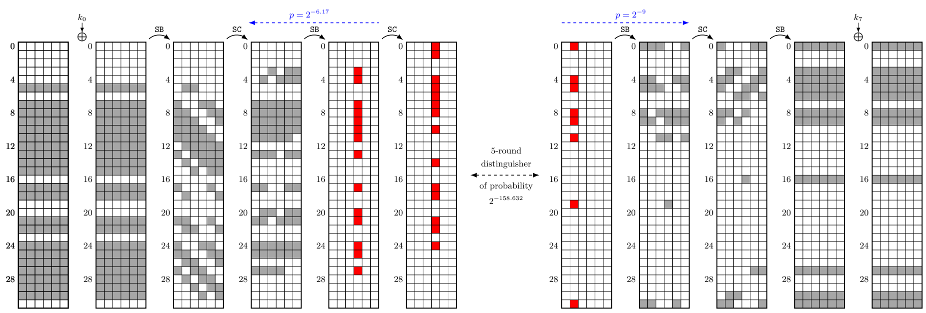

Fig. 2: Attack against `SPEEDY-7-192`.

The attack starts by the generation of pairs. We need $2^{173.80}$ pairs to get one following the differential and the probabilistic extension. This is done in a classical way by asking for several structures of plaintexts and sieving the possible pairs using a hash table indexed by the inactive lines of the ciphertext. The data complexity is thus $2^{173.80+1} = 2^{174.80}$ chosen plaintexts.

We then used the KyRyDi tool [2] to search for an efficient key recovery procedure, and manually verified its output to ensure the correctness of the results. Candidate pairs are first filtered based on the possible differences at the input of the first S-box layer and at the output of the last one. The filters for each row are provided in Table 1.

The second step of the attack consists, for each remaining pair, in recovering the values of the active rows for the first two and the last two S-box layers. This is achieved through a back-and-forth procedure between the four involved S-box layers, exploiting the key schedule to reduce the overall complexity. As a result, the KyRyDi tool identified a strategy with an overall cost of $2^{173.80+0.41} = 2^{174.21}$ simple operations to process the $2^{173.80}$ pairs and outputs $2^{173.80-29.17} = 2^{144.63}$ tuples composed of a pair and a key candidate. The procedure found by KyRyDi is depicted in Figure 3. Each of them suggests a value for 192 bits of k_0 and k_7. However, 47 of these bits are shared between k_0 and k_7, so the total entropy of the recovered key material is only 145 bits. These key bits are highlighted in red in Table 2.

First-round S-boxes				Last-round S-boxes	
Row	Filter	Row	Filter	Row	Filter
5	0.476	18	0.476	0	0.093
7	0.069	21	0.117	3	0.046
8	0.023	22	0.046	4	0
9	0	24	0.069	5	0.023
10	0	25	0.023	6	0.046
11	0	26	0.046	8	0.069
12	0	27	0.069	9	0.752
13	0	28	0.117	16	1.541
14	0	29	0.167	27	0.956
15	0.167	30	0.642	30	0.752
17	0.385			31	0.046
Total filter: 7.216					

Table 1: Filtering (in $\log_2$) in the active S-boxes of the first S-box and the last S-box application during the attack against `SPEEDY-7-192`.

We still need to recover 47 bits to reconstruct the full master key. Brute-forcing these bits would result in a total complexity of approximately $2^{144.63+47} = 2^{191.63}$, which is close to the brute-force bound. One option would be to increase the number of structures in order to apply more filtering and reduce the number of key candidates. But the filter per structure is really small and a more efficient approach is to reuse the strategy we applied in [3].

Indeed, during the second step of the attack, we recovered the values of all the active rows. This implies that there are several inactive rows in both the plaintext and ciphertext for which we know at least one bit after the first S-box layer or before the last one. By combining this additional information with the key bits already recovered (highlighted in blue in Table 2), we proceed as follows: we first guess the missing bits for rows 2, 7, 11, 18 and 29 of the ciphertext, as these rows offer additional filtering. Then we guess the bits for rows 1, 10, 14, 15, 17, 19, 26 and 28 which yield more key bits than their guessing cost. At this point, we are left with $2^{144.63+6\times13-18-46} = 2^{158.63}$ candidates for $145 + 6 \times 13 - 46 = 177$ key bits. We can then perform the same procedure for rows 0, 4, 6, 16, 19, 20, 23 and 31 of the plaintext and guess the missing key bits.

Hence, the overall time complexity is dominated by the data processing phase required to generate the necessary pairs, resulting in a total time complexity of $2^{174.80}$. The memory complexity is determined by the need to store the required structures built from the ciphertexts, which amounts to $2^{6\times11} = 2^{66}$.

References

1. Beyne, T., Neyt, A.: Improved differential cryptanalysis of SPEEDY. Cryptology ePrint Archive, Paper 2025/915 (2025), https://eprint.iacr.org/2025/915

Row number	Master key bits						# guessed
0	161	168	175	182	189	4	4
1	11	18	25	32	39	46	2
2	53	60	67	74	81	88	6
3	95	102	109	116	123	130	4
4	137	144	151	158	165	172	6
5	179	186	1	8	15	22	1
6	29	36	43	50	57	64	4
7	71	78	85	92	99	106	5
8	113	120	127	134	141	148	4
9	155	162	169	176	183	190	5
10	5	12	19	26	33	40	1
11	47	54	61	68	75	82	6
14	173	180	187	2	9	16	2
15	23	30	37	44	51	58	4
16	65	72	79	86	93	100	5
17	107	114	121	128	135	142	3
18	149	156	163	170	177	184	6
19	191	6	13	20	27	34	1
26	101	108	115	122	129	136	3
27	143	150	157	164	171	178	5
28	185	0	7	14	21	28	1
29	35	42	49	56	63	70	6
30	77	84	91	98	105	112	5
31	119	126	133	140	147	154	4

Table 2: Key bits of the round key k_7 that are involved in the attack against full `SPEEDY-7-192`. Coloured bits are also involved in k_0 and blue ones are known at the end of the second step of the key recovery.

2. Boura, C., David, N., Derbez, P., Boissier, R.H., Naya-Plasencia, M.: A generic algorithm for efficient key recovery in differential attacks - and its associated tool. In: Joye, M., Leander, G. (eds.) EUROCRYPT 2024, Part I. LNCS, vol. 14651, pp. 217–248. Springer, Cham (May 2024). `https://doi.org/10.1007/978-3-031-58716-0_8`
3. Boura, C., Derbez, P., Germon, B., Boissier, R.H., Naya-Plasencia, M.: SPEEDY: Caught at last. Cryptology ePrint Archive, Paper 2025/890 (2025), `https://eprint.iacr.org/2025/890`

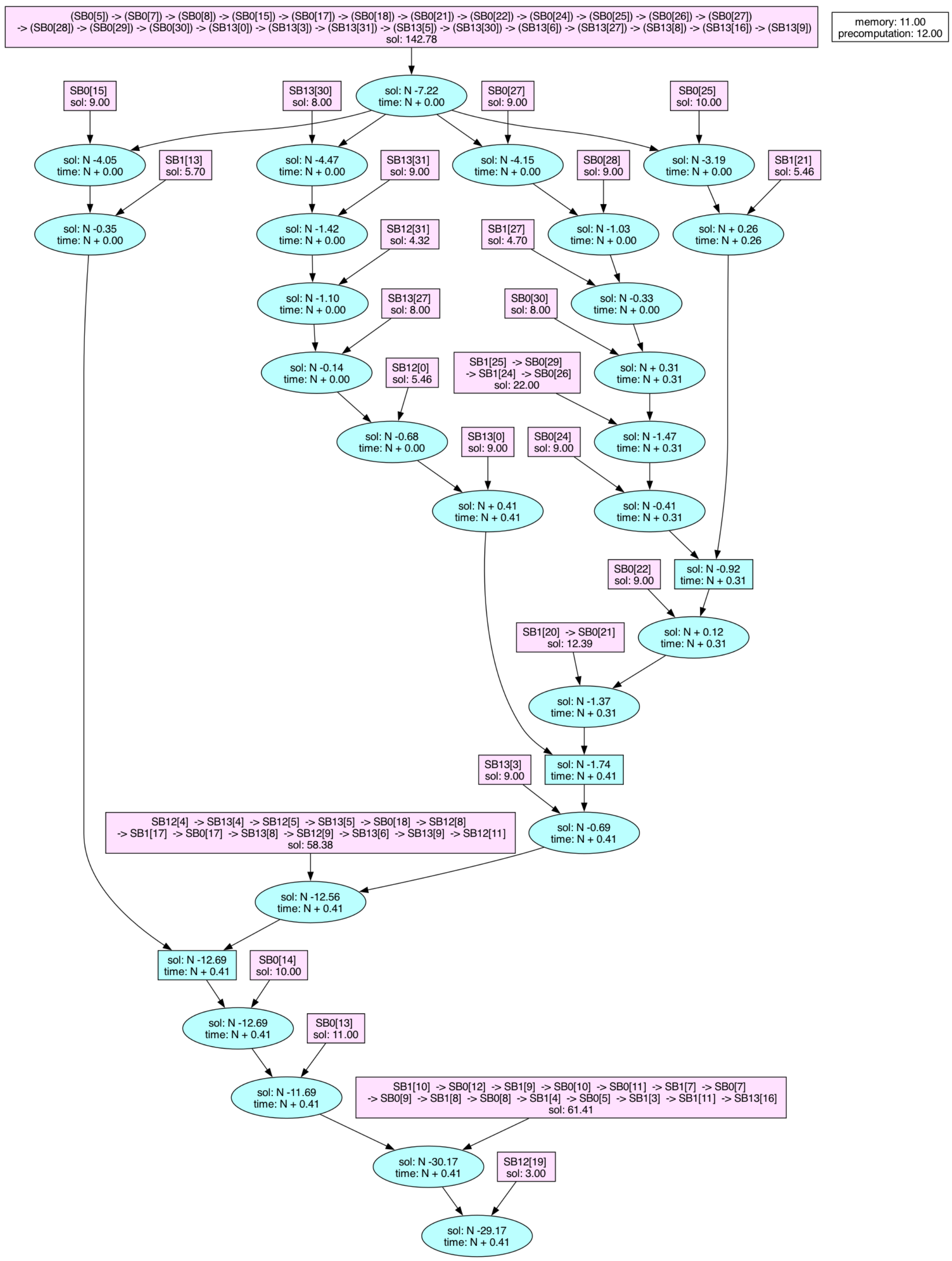

Fig. 3: Graph of the key-recovery of our attack against `SPEEDY-7-192`